

Pasan Kalansooriya

PhD Student - University of Melbourne

+61449923304 pasanxavioz@gmail.com [GitHub](#)

Unit 504, 518 Swanston Street, Carlton VIC 3053

[LinkedIn](#)

[Medium Blogs](#)

I am a Computer Science and Engineering graduate, currently pursuing a PhD at the University of Melbourne, with a strong foundation in software engineering and research. My core interests lie in Computer Vision and Natural Language Processing, with a proven track record of applying machine learning and deep learning techniques to solve real-world problems. With international research experience in Japan, industry training at WSO2 and Sysco LABS, and a deep commitment to continuous learning, I aim to contribute to impactful research and the development of intelligent systems that bridge the gap between academia and industry.

EDUCATION & EXTRACURRICULAR ACTIVITIES

University of Melbourne - Australia (May 2025 - Present)

PhD – Faculty of Medicine, Dentistry and Health Science

- Awarded the competitive, merit-based ‘Melbourne Research Scholarship’, fully funding my PhD research
- Project: A Decision Support System for the Treatment of Vestibular Schwannoma
- Societies: Event Organizer (2025-2026) - Sri Lankan Graduates’ Society

University of Moratuwa - Sri Lanka (2019 - 2024)

BSc Eng. (Hons) Computer Science and Engineering

- **GPA: 4.04** (out of 4.2)
- Achievements:
 - Dean's List for all the semesters
 - Selected as the **only student from Sri Lanka** to participate in [Science Tokyo AOTULE](#) (Asia-Oceania Top University League on Engineering) Summer Program 2024
 - Presented the research and chaired a session at the 15th Multidisciplinary International Student Workshop (MISW), Institute of Science Tokyo
- Extra-curricular activities
 - Member of the Badminton team
 - UMiSF (University of Moratuwa International Shuttlers’ Fest) 2020 – Organizing Committee Member
 - Carrier fair of CSE department 2021 – Company Coordinator
 - EXMO (2023) University of Moratuwa – Department Facilitator

CIMA - The Chartered Institute of Management Accountants (2019)

Certificate in Business Accounting (Cert BA)

- Completed the Certificate-level

S. Thomas’ College, Mt. Lavinia – Sri Lanka (2016 – 2018)

- G.C.E Advanced Level Examination 2018: Four distinction passes with a z-score of 2.1114 (All-Island Rank: 326)
- The Professor Dayasiri Fernando Memorial Prize Winner – 2019
- Served as a Teaching Assistant (2019 Jan – 2019 Jun)
- Societies: Association of Science and Technology | Society for Social Work and Welfare

EXPERIENCE

Casual Research Assistant at La Trobe University - Bundoora, Melbourne

From 18th March 2026 - Present

- Collaborative research project with the University of Auckland (NZ) and La Trobe University.
- Project: Multi-Site Accelerometry Validation. Tasks:
 - Developed an end-to-end R-based pipeline to process raw 100 Hz triaxial accelerometer data, collected directly by the project stakeholders, into actionable clinical activity insights.
 - Automated feature extraction and data filtering (for sleep and non-wear) to ensure data integrity, ultimately generating participant summaries that detailed activity intensity by epoch count, percentage, and duration.
 - Implemented several published adolescent-specific models, including Artificial Neural Networks (ANN) and regression equations, to benchmark classification accuracy.

Software Engineer at Sysco LABS

From 9th September 2024 - 18th April 2025 (8 months)

- Sysco LABS is the captive innovation arm of Sysco Corporation (NYSE: SYY), a Fortune 500 company and the world’s largest food service company.
- My Tasks:
 - Participated in the 'Project Summit' training program, developing an end-to-end web application mimicking the Sysco-shop (a customer food ordering platform). This project utilized React, Java (Spring Boot), NodeJS, MongoDB, and AWS.
 - Assisted with tasks on Merchandising and Supply Chain projects for the Sysco-Canada Team.

Tokyo Tech AOTULE Summer Program 2024 - Funakoshi Lab - Tokyo Institute of Technology, Tokyo, Japan

From 1st June 2024 - 31st Aug 2024 (3 months)

- Research Focus: Multimodal Dialogue Systems
- Project: Turn-taking Prediction in Conversational AI
 - Conducted literature review on turn-taking mechanisms, including integration of physiological signals.
 - Utilized a dataset of 96 Japanese dialogue sessions (each 15 minutes) involving 48 speaker pairs to predict turn-taking behavior.
 - Incorporated methodologies from notable works, including Voice Activity Projection (VAP) for self-supervised learning.
- Presented my home university research and chaired a session at the 15th Multidisciplinary International Students Workshop (MISW) at the Tokyo Institute of Technology.

Visiting Instructor - Department of Computer Science and Engineering, University of Moratuwa

From 24th July 2023 - 03rd Nov 2023 (3 months) & From 22nd Jan 2024 - 31st May 2024 (4 months)

- For modules:
 - CS2813 Visual Programming
 - CS1033 Programming Fundamentals
- Tasks:
 - Conducted group sessions to assist students in need.
 - Contributed to the question banks by generating innovative questions and obtaining approvals after discussions with lecturers.
 - Assisted lecturers in marking lab work for the modules

Software Engineer Intern at WSO2

From 2nd January 2023 - 30th June 2023 (6 months)

- Project: Migrate existing “Updates delivery subsystem to Azure”
- Tasks:
 - Research multiple architectural solutions to implement the migration with several Azure services (Azure API management, Azure CDN, Azure Blob storage etc.)
 - Demonstrate the possible solutions’ pros and cons to team leads and get approval for the most preferred approach.
 - Use Azure SDK for Java/Python to change existing codebases to support the migration.
 - Develop scripts to migrate currently deployed artifacts (on Nexus and SVN repositories) to Azure blob storage.

SKILLS

Programming Languages	Java, Python, JavaScript, Typescript, PHP
Databases	MySQL, MongoDB, PostgreSQL
Backend	Node.js, Express.js, Spring Boot
Frontend	React, Bootstrap, Single-SPA
Version Controlling	Git, GitHub
Other	Elastic-search, AWS, Azure
Non-technical	Teamwork and communication, Project management, Leadership, and Ability to adapt.

PUBLICATIONS

P. Kalansooriya, S. Wijewickrema, S. Kandanaarachchi, B. Copson, J.-M. Gerard, and S. O’leary, "Machine learning for preoperative predictions in Vestibular Schwannoma: A systematic review," *Neuroscience Informatics*, vol. 6, no. 2, p. 100273, June 2026.

- **Q1 journal** (Elsevier)
- PhD first publication - Systematic Literature Review
- Available at: <https://www.sciencedirect.com/science/article/pii/S2772528626000178>

K. Chavinda, P. Kalansooriya, T. Weerasuriya, N. de Silva, U. Thayasivam, K. Srikandabala, K. Prabagar, and D. Alahakoon, "An Emotion Aware Context Adaptive Machine Learning Approach for Detecting Hate Speech in Social Media," *Neural Computing and Applications*, 2026

- Undergraduate Final Year Project (University of Moratuwa)
- **Q1 journal** (Springer Nature) **ranked #10** globally in Google Scholar Metrics for Artificial Intelligence
- Available online: <https://link.springer.com/article/10.1007/s00521-026-12029-8>

PROJECTS

Unmasking Hate: An Integrated Approach to Detecting Hate Speech in Social Media

Final Year project

- A collaborative project with the **Centre for Data Analytics and Cognition (CDAC) - La Trobe University**
- Developed and deployed a novel approach to detect hate speech in online content (tweets) that uses dual contrastive learning, emotion modeling, and author profiling techniques. Surpassed the performance of the baseline Dual Contrastive Learning (DCL) framework
- Obtained A+ for the final year project module
- [GitHub Repository](#) | [Deployment](#)

AudioMNIST classification

Semester 7 Machine Learning Module's Project | Individual Project

- Engaged in a machine learning classification project, focusing on feature engineering, hyperparameter tuning, and implementing explainable AI for enhanced model interpretability.
- Technologies: scikit-learn, Lime, pandas
- [GitHub Repository](#)

Sinhala Poems' Metaphors Search Application

Semester 7 Information Retrieval Module's Project | 2023 Sept – 2023 Oct | Individual Project

- A search application powered by elastic search for searching Sinhala metaphors in a corpus. Search can be based on the source domain, target domain, poet, and year of the metaphor.
- Technologies: Elastic search, kibana, flask
- [GitHub Repository](#)

Workflow Management System

Semester 5 Software Engineering Module's Project

- A proof of concept for Galle district Irrigation department. First experience of working with a real client.
- The DI (Director of Irrigation) user can create a workflow with ordered subtasks and assign employees in the system to each subtask. Then the employees can carry out the work specified in the subtask and complete it. Assigned users and DI can view the current progress and each subtask completion, deletion, and other special events will be notified to the relevant users.
- Technologies used: React with MaterialUI, ExpressJs, NodeJs, and MongoDB
- GitHub Repository: [backend](#) | [frontend](#)

Airline reservation System

Semester 4 Database System Module's Project

- The system allows customers of the airline company to see flight details, make flight reservations get registered to the system to be eligible to get discounts based on the frequency of booking tickets, etc. Also, the management of the airline company can use this system to Create/update flight schedules, create/update aircraft details in the company database, generate reports (based on the passenger details and flight details got saved in the database), etc.
 - Technologies used: React with bootstrap, ExpressJs, NodeJs, and MySQL
 - [GitHub Repository](#)
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CERTIFICATES & ACHIEVEMENTS

Microsoft Certified: Azure Fundamentals

Microsoft | 2023 January - [Credential](#)

AI/ML Engineer – Stage 1

SLIIT - Centre for Open and Distance Education |
April 2023 - [Credential](#)

AI/ML Engineer – Stage 2

SLIIT - Centre for Open and Distance Education |
April 2023 - [Credential](#)

Machine Learning

Coursera – Stanford Online | 2021 Aug - [Credential](#)

Neural Networks and Deep Learning

Coursera – DeepLearning.AI | 2021 Oct - [Credential](#)

Introduction to Cybersecurity

Cisco | 2021 Aug - [Credential](#)

Hackerrank certificates and achievements - [Profile](#)

CyberZee'23 Inter-University Cyber-Security Quiz Competition - Semi-Finalist

Issued by University of Kelaniya ISACA Student Group and ISACA Sri Lanka Chapter · Jul 2023

NON-RELATED REFEREES

A/Prof Sudanthi Wijewickrema
Honorary Principal Research Fellow
Department of Surgery
University of Melbourne.
Australia

Mobile: +61431084939
Email: sudanthi.wijewickrema@unimelb.edu.au

Sevvandi Kandanaarachchi PhD
Senior Research Scientist
CSIRO's Data61
Wurundjeri land
Gate 5, Normanby Road, Clayton VIC 3168

Mobile: +61490206018
Email: sevvandi.Kandanaarachchi@data61.csiro.au